

# CLIMATE AND THERMAL MANAGEMENT

## High-Level Design Document

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## 1. System Overview

The Climate and Thermal Management coordinates its functions through typed AUTOSAR interfaces and ports. Software components exchange signals through provider and consumer ports.

## 2. Software Architecture

The primary software components are ClimateManager, CompressorControl, BlowerControl, CabinSensorManager, CoolantControl, and DefrostManager.

## 3. Software Components

| Component          | Role        | Primary Responsibility                  |
|--------------------|-------------|-----------------------------------------|
| ClimateManager     | Coordinator | Computes cabin temperature targets.     |
| CompressorControl  | Consumer    | Controls the A/C compressor.            |
| BlowerControl      | Consumer    | Controls blower fan speed.              |
| CabinSensorManager | Provider    | Measures cabin and outside temperature. |
| CoolantControl     | Provider    | Reports coolant circuit state.          |
| DefrostManager     | Consumer    | Activates window defrost.               |

## 4. Interfaces

| Interface           | Provider           | Consumer          | Signals            |
|---------------------|--------------------|-------------------|--------------------|
| ICabinTemperature   | CabinSensorManager | ClimateManager    | CabinTemperature   |
| IOutsideTemperature | CabinSensorManager | ClimateManager    | OutsideTemperature |
| ICompressorRequest  | ClimateManager     | CompressorControl | CompressorSpeed    |
| IBlowerRequest      | ClimateManager     | BlowerControl     | BlowerSpeed        |
| ICoolantState       | CoolantControl     | ClimateManager    | CoolantStatus      |
| IDefrostRequest     | ClimateManager     | DefrostManager    | DefrostCommand     |

## 5. Ports

| Component          | Port                   | Direction | Interface           |
|--------------------|------------------------|-----------|---------------------|
| CabinSensorManager | CabinTemperature_Out   | P-PORT    | ICabinTemperature   |
| ClimateManager     | CabinTemperature_In    | R-PORT    | ICabinTemperature   |
| CabinSensorManager | OutsideTemperature_Out | P-PORT    | IOutsideTemperature |
| ClimateManager     | OutsideTemperature_In  | R-PORT    | IOutsideTemperature |
| ClimateManager     | CompressorRequest_Out  | P-PORT    | ICompressorRequest  |
| CompressorControl  | CompressorRequest_In   | R-PORT    | ICompressorRequest  |
| ClimateManager     | BlowerRequest_Out      | P-PORT    | IBlowerRequest      |
| BlowerControl      | BlowerRequest_In       | R-PORT    | IBlowerRequest      |
| CoolantControl     | CoolantState_Out       | P-PORT    | ICoolantState       |
| ClimateManager     | CoolantState_In        | R-PORT    | ICoolantState       |
| ClimateManager     | DefrostRequest_Out     | P-PORT    | IDefrostRequest     |
| DefrostManager     | DefrostRequest_In      | R-PORT    | IDefrostRequest     |

## 6. Signals

| Signal             | Data Type | Source             | Destination       |
|--------------------|-----------|--------------------|-------------------|
| CabinTemperature   | int16     | CabinSensorManager | ClimateManager    |
| OutsideTemperature | int16     | CabinSensorManager | ClimateManager    |
| CompressorSpeed    | uint16    | ClimateManager     | CompressorControl |
| BlowerSpeed        | uint8     | ClimateManager     | BlowerControl     |
| CoolantStatus      | uint8     | CoolantControl     | ClimateManager    |
| DefrostCommand     | uint8     | ClimateManager     | DefrostManager    |

## 7. Dependencies

| Source             | Relationship       | Target              | Reason                         |
|--------------------|--------------------|---------------------|--------------------------------|
| CabinSensorManager | PROVIDES_INTERFACE | ICabinTemperature   | Publishes cabin temperature.   |
| ClimateManager     | REQUIRES_INTERFACE | ICabinTemperature   | Consumes cabin temperature.    |
| CabinSensorManager | PROVIDES_INTERFACE | IOutsideTemperature | Publishes outside temperature. |
| ClimateManager     | REQUIRES_INTERFACE | IOutsideTemperature | Consumes outside temperature.  |
| ClimateManager     | PROVIDES_INTERFACE | ICompressorRequest  | Publishes compressor speed.    |
| CompressorControl  | REQUIRES_INTERFACE | ICompressorRequest  | Consumes compressor speed.     |
| ClimateManager     | PROVIDES_INTERFACE | IBlowerRequest      | Publishes blower speed.        |
| BlowerControl      | REQUIRES_INTERFACE | IBlowerRequest      | Consumes blower speed.         |
| CoolantControl     | PROVIDES_INTERFACE | ICoolantState       | Publishes coolant status.      |
| ClimateManager     | REQUIRES_INTERFACE | ICoolantState       | Consumes coolant status.       |
| ClimateManager     | PROVIDES_INTERFACE | IDefrostRequest     | Publishes defrost command.     |
| DefrostManager     | REQUIRES_INTERFACE | IDefrostRequest     | Consumes defrost command.      |

## 8. Functional Flows

### Flow F-001: Cabin Temperature Distribution

CabinSensorManager publishes CabinTemperature through ICabinTemperature. ClimateManager receives the signal through CabinTemperature\_In and updates its internal state model.

### Flow F-002: Outside Temperature Distribution

CabinSensorManager publishes OutsideTemperature through IOutsideTemperature. ClimateManager receives the signal through OutsideTemperature\_In and updates its internal state model.

### Flow F-003: Compressor Speed Distribution

ClimateManager publishes CompressorSpeed through ICompressorRequest. CompressorControl receives the signal through CompressorRequest\_In and updates its internal state model.

### Flow F-004: Blower Speed Distribution

ClimateManager publishes BlowerSpeed through IBlowerRequest. BlowerControl receives the signal through BlowerRequest\_In and updates its internal state model.

### Flow F-005: Coolant Status Distribution

CoolantControl publishes CoolantStatus through ICoolantState. ClimateManager receives the signal through CoolantState\_In and updates its internal state model.

### Flow F-006: Defrost Command Distribution

ClimateManager publishes DefrostCommand through IDefrostRequest. DefrostManager receives the signal through DefrostRequest\_In and updates its internal state model.

## 9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

## **10. Engineering Traceability**

This synthetic HLD supports extraction, retrieval, validation and revision comparison experiments.